

# Stereotactic Radiosurgery of Large Uveal Melanomas with the Gamma-knife

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**Objective:** To present our experience with the Gamma-knife in treating large uveal melanomas with stereotactic radiosurgery.

**Design:** Prospective, noncomparative, interventional case series.

**Participants:** Fifty-eight patients with unilateral uveal melanomas were treated from 1996 through 1999 with stereotactic radiosurgery using the Gamma-knife. From these we report the results of 35 patients who had a follow-up of more than 1 year after irradiation.

**Intervention:** Stereotactic radiosurgery with the Gamma-knife.

**Main Outcome Measures:** Tumor control, maximum apical tumor height, eye retention rate, and visual acuity.

**Results:** In 34 eyes (97%), local tumor control was achieved. The maximum apical tumor height decreased from a median of 9.1 mm (95% confidence interval [CI], 3.2–13.9 mm) before treatment to 6.2 mm (95% CI, 2.1–11.9 mm) at 1 year after treatment ( $P < 0.001$ , paired  $t$  test). The tumor volume decreased from a median of 0.8 cm<sup>3</sup> before treatment to 0.5 cm<sup>3</sup> 1 year after treatment ( $P < 0.001$ , paired  $t$  test). Two eyes required enucleation (one radiation failure, one secondary glaucoma). The median visual acuity decreased from 20/60 (95% CI, hand movement [HM] to 20/20) before treatment to 20/200 (95% CI, HM to 20/30) at 1 year after treatment ( $P = 0.001$ , paired  $t$  test).

**Conclusions:** Stereotactic radiosurgery using the Gamma-knife is an alternative to enucleation in treating large uveal melanomas. The visual function may be preserved in selected cases. *Ophthalmology* 2000;107:1381–1388 © 2000 by the American Academy of Ophthalmology.

To date, a variety of therapeutic options exist for the treatment of uveal melanomas. Small- and medium-sized tumors are treated mainly with different forms of radiation or laser therapy.<sup>1–6</sup> In contrast, the treatment of large uveal melanomas is more difficult and thus remains controversial. Surgical approaches such as enucleation are still advocated by many ophthalmologists for large uveal melanomas, especially if useful vision cannot be salvaged.<sup>7</sup> The efficacy of transscleral local resection of large uveal melanomas has been demonstrated, but the procedure has not gained wide acceptance so far, mostly because of concern about the potential risks of the systemic arterial hypotension that is required during surgery.<sup>8–10</sup> In 1977, Gragoudas et al<sup>11</sup> introduced proton beam therapy for the treatment of small uveal melanomas. The same authors have also demonstrated that proton beam therapy provides an acceptable alternative to enucleation for large uveal melanomas.<sup>2,12,13</sup> Others have confirmed that large uveal melanomas may be successfully

treated by irradiation using charged particles such as proton beams or helium ions.<sup>14–21</sup> However, both forms of charged particle radiation therapy depend on the availability of large-scale facilities, which are extremely expensive and scarce.

Stereotactic radiosurgery may be an alternative to charged particle therapy for the treatment of large uveal melanomas. During the 1960s, the concept of stereotactic radiosurgery was envisioned by the Swedish neurosurgeon Lars Leksell, and the first Gamma-knife unit was developed shortly afterwards together with the radiobiologist Borje Larsson. Stereotactic radiosurgery usually consists of a single-session delivery of ionizing radiation to a stereotactically localized volume of tissue. This form of therapy is currently gaining popularity in Europe as an alternative to neurosurgery for cranial tumors and arteriovenous malformations.<sup>22–26</sup>

The treatment of uveal melanomas with the Gamma-knife has been reported before. However, these treatments are either experimental,<sup>27</sup> or the patient number is relatively small,<sup>28,29</sup> or the ionizing radiation is applied fractionated,<sup>30</sup> or the treatment parameters were varied within the reported groups.<sup>31,32</sup> A Leksell Gamma-knife has been available to us in Munich since 1995 (Fig 1). In this report, we describe our experience in treating consecutively all large and unfavorably located uveal melanomas as an alternative to enucleation with standardized treatment parameters in a (non-fractionated) single treatment procedure.

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